t2-w09 — condition C3

Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

> C2 plus a validation and repair pass over the draft.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

```
Mathematics
```

`grade_level`

```
JSS1
```

`topic`

```
Simplification of Algebraic Terms
```

`subtopic`

```
Simplification of algebraic expressions: order of operations, brackets and word problems
```

`term`

```
Second Term
```

`week`

```
9
```

`lesson_objectives`

- carry out order of operation in algebra
- insert/remove bracket and simplify expression
- solve word problems involving algebraic expression

entry_behaviour

- Addition and subtraction of algebraic terms
- Multiplication and division of algebraic terms

instructional_materials

- Chart showing algebraic expression.

Objective 1 — carry out order of operation in algebra

objective_text

carry out order of operation in algebra

current_step

1

instructional_events

- provide_learning_guidance
- elicit_performance
- provide_feedback

student_tasks

- 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

assessment_points

- Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
- Simplify $7 \times 3a - (3a + 5a) \times 2$

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Example 1. Find the value of the following\n\n(a) $18 − 6 \\times 2$\n\n(b) $6
− 18 \\div 3$\n\n(c) $6 \\times 2 − 2 − 40 \\div 4$\n\n**Solution**\n\n(a) $18 − 6 \\times 2$
\n\n$= 18 − 12$\n\n$= 6$\n\n(b) $6 − 18 \\div 3$\n\n$= 6 − 6$\n\n$= 0$\n\n(c) $(6 \\times 2) − 2
− (40 \\div 4)$\n\n$= 12 − 2 − 10$\n\n$= 10 − 10$\n\n$= 0$\n\n#### Example 2. Simplify the
following\n\n(a) $3x \\times 2 + 5x$\n\n(b) $17y − 5 \\times 2y$\n\n**Solution**\n\n(a) $3x \
\times 2 + 5x$\n\n$= 6x + 5x$\n\n$= 11x$\n\n(b) $17y − 5 \\times 2y$\n\n$= 17y − 10y$\n\n$= 7y$
\n\n#### Example 3. Simplify the following\n\n(a) $7x \\div x + 5$\n\n(b) $4x − (6 \\div 3)$
\n\n(c) $(24x \\div 6) + x + (1 \\times 5x)$\n\n**Solution**\n\n(a) $7x \\div x + 5$\n\n$= \
\frac{7x}{x} + 5$\n\n$= 7 + 5$\n\n$= 12$\n\n(b) $4x − (6 \\div 3)$\n\n$= 4x − 2$\n\n(c) $(24x \
\div 6) + x + (1 \\times 5x)$\n\n$= 4x + x + 5x$\n\n$= 5x + 5x$\n\n$= 10x$\n\n#### Activity.
Simplify the following\n\n1. $4n + 3n \\times 10$\n2. $32 + 8x \\div 2$\n3. $3 \\times 5x + 4x \
\div 2$\n4. $36y \\div 6 + y + 2 \\times 4y$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — insert/remove bracket and simplify expression

objective_text

insert/remove bracket and simplify expression

current_step

2

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

assessment_points

• Simplify $12 - (7 - 2f) - (3 - f)$
• Simplify $(8a - 5) + (14 - 3a)$

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Hints**\n\n- Always try to simplify the terms inside bracket first.\n- A $+$ sign
before a bracket does not change the signs of the terms inside the bracket, when\n  bracket is
removed.\n- A $-$ sign before a bracket will change all the $+$ sign inside the bracket to $-$
and also\n  change all the $-$ sign inside the bracket to $+$ when the bracket is removed.
\n\n#### Example 1. Simplify\n\n(a) $a + (b + c)$\n\n(b) $a + (b - c)$\n\n(c) $2a - (3b - c)$
\n\n**Solution**\n\n(a) $a + (b + c) = a + b + c$\n\n(b) $a + (b - c) = a + b - c$\n\n(c) $2a -
(3b - c) = 2a - 3b + c$\n\n#### Example 2. Simplify\n\n(a) $a + (a - b)$\n\n(b) $6x + (3 - x)$
\n\n(c) $(3a + 2) + (5a + a)$\n\n**Solution**\n\n(a) $a + (a - b)$\n\n$= a + a - b$\n\n$= 2a - b$
\n\n(b) $6x + (3 - x)$\n\n$= 6x + 3 - x$\n\n$= 6x - x + 3$\n\n$= 5x + 3$\n\n(c) $(3a + 2) + (5a +
a)$\n\n$= 3a + 2 + 5a + a$\n\n$= 3a + 5a + a + 2$\n\n$= 9a + 2$\n\n#### Example 3. Simplify the
following\n\n(a) $5x - (3y + z)$\n\n(b) $10d - (9c - 4d)$\n\n(c) $(6x - y) - (4x - 5y)$
\n\n**Solution**\n\n(a) $5x - (3y + z)$\n\n$= 5x - 3y - z$\n\n(b) $10d - (9c - 4d)$\n\n$= 10d -
9c + 4d$\n\n$= 10d + 4d - 9c$\n\n$= 14d - 9c$\n\n(c) $(6x - y) - (4x - 5y)$\n\n$= 6x - y - 4x +
5y$\n\n$= 6x - 4x + 5y - y$\n\n$= 2x + 4y$\n\n$= 2(x + 2y)$\n\n#### Students activity\n\n(a)
$(12a - 7) + (4 - 5a)$\n\n(b) $(8x - 2y) - (9x - 4y)$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — solve word problems involving algebraic expression

objective_text

solve word problems involving algebraic expression

current_step

3

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

assessment_points

• A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Example 1\n\nA pen cost $P$ naira and a pencil cost 15 naira less than a pen.
\n\n(a) Find the cost of a pencil\n\n(b) Find the cost of a pen and a
pencil\n\n**Solution**\n\nCost of a pen $= ₦P$\n\n(a) cost of a pencil $= ₦(P − 15)$\n\n(b) So
cost of a pen and a pencil $= ₦P + ₦(P − 15)$\n\n$= ₦P + ₦P − ₦15$\n\n$= ₦2P − ₦15$\n\n$= ₦(2P −
15)$\n\n#### Example 2\n\nA pencil cost $x$ naira and a rubber cost 40 naira more than a pencil.
\n\n(a) find the cost of a rubber\n\n(b) find the cost of a pencil and a
rubber\n\n**Solution**\n\n(a) cost of a pencil $= ₦x$\n\n$\\therefore$ cost of a rubber $= ₦(x +
40)$\n\n(b) cost of a pencil and a rubber:\n\n$= ₦x + ₦(x + 40)$\n\n$= ₦(x + x + 40)$\n\n$= ₦(2x
+ 40)$\n\n#### Example 3\n\nIn a class, there are $x$ boys and $y$ girls last semester. This
semester, 5 boys join and\n3 girls leave the class. How many students are now in this class?
\n\n**Solution**\n\nTotal no $=$ number of boys $+$ number of girls\n\nNumber of boys $= x + 5$
\n\nNumber of girls $= y − 3$\n\nTotal number $= x + 5 + y − 3$\n\n$= x + y + 5 − 3$\n\n$= x + y
+ 2$\n\n#### Student activity\n\n1. A pen cost $h$ naira and a crayon cost ₦20 less than a pen.
\n   (a) Find the cost of a crayon\n   (b) Find the cost of a crayon and a pen\n2. Envelopes cost
$d$ naira per pack, 4 note pad cost 20 naira more than a pack of envelopes.\n   Find the total
cost of a pack of envelope and notepad."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Simplifying Algebraic Expressions – Order of Operations

EXPLANATION

When you simplify an algebraic expression you must follow the same order of operations you use with numbers:

1. **Brackets** – do anything inside brackets first.
2. **Multiplication and Division** – work from left to right.
3. **Addition and Subtraction** – work from left to right.

Remember that multiplication and division are of equal priority, as are addition and subtraction. Always treat the symbols \(\times\) and \(\div\) the same way, moving left-to-right.

For example, in the expression \(4n + 3n \times 10\) you first multiply \(3n\) by \(10\) to get \(30n\), then add \(4n\) to obtain \(34n\).

WORKED_EXAMPLE

Worked Example 1

Simplify \(18 - 6 \times 2\)

- First do the multiplication: \(6 \times 2 = 12\).
- Then subtraction: \(18 - 12 = 6\).

Answer: \(6\)

Worked Example 2

Simplify \(6 - 18 \div 3\)

- Division first: \(18 \div 3 = 6\).
- Subtraction: \(6 - 6 = 0\).

Answer: \(0\)

Worked Example 3

Simplify \(6 \times 2 - 2 - 40 \div 4\)

- Multiplication: \(6 \times 2 = 12\).
- Division: \(40 \div 4 = 10\).
- Now work left-to-right with addition/subtraction: \(12 - 2 = 10\), then \(10 - 10 = 0\).

Answer: \(0\)

WORKED_EXAMPLE

Worked Example 4

Simplify \(3x \times 2 + 5x\)

- Multiply first: \(3x \times 2 = 6x\).
- Then add: \(6x + 5x = 11x\).

Answer: \(11x\)

Worked Example 5

Simplify \(17y - 5 \times 2y\)

- Multiply: \(5 \times 2y = 10y\).
- Subtract: \(17y - 10y = 7y\).

Answer: \(7y\)

Worked Example 6
Simplify \(7x \div x + 5\)

- Division: \(7x \div x = 7\) (since \(x\) cancels).
- Add: \(7 + 5 = 12\).

Answer: \(12\)

Worked Example 7
Simplify \(4x - (6 \div 3)\)

- Inside brackets: \(6 \div 3 = 2\).
- Subtract: \(4x - 2\) (cannot simplify further without a value for \(x\)).

Answer: \(4x - 2\)

Worked Example 8
Simplify \((24x \div 6) + x + (1 \times 5x)\)

- First division: \(24x \div 6 = 4x\).
- Multiplication: \(1 \times 5x = 5x\).
- Combine like terms: \(4x + x + 5x = 10x\).

Answer: \(10x\)

1. 1. Simplify \(4n + 3n \times 10\)
   answer: 34n
2. 2. Simplify \(32 + 8x \div 2\)
   answer: 32 + 4x
3. 3. Simplify \(3 \times 5x + 4x \div 2\)
   answer: 15x + 2x
4. 4. Simplify \(36y \div 6 + y + 2 \times 4y\)
   answer: 6y + y + 8y = 15y
5. 5. Simplify \(14pq \times r\)
   answer: 14pqr
6. 6. Simplify \(12x^2y \div 3x\)
   answer: 4xy
7. 7. Simplify \(7 \times 3a - (3a + 5a) \times 2\)
   answer: 21a - 16a = 5a

Removing Brackets and Simplifying Algebraic Expressions

EXPLANATION

When you remove brackets, pay attention to the sign that is just before the opening bracket.

* A **plus (+) sign** in front of a bracket does **not** change any of the signs inside the bracket. You can simply drop the brackets and keep the terms as they are.
* A **minus (–) sign** in front of a bracket **reverses** every sign inside the bracket: each '+' becomes '–' and each '–' becomes '+'.

Steps to simplify an expression with brackets
1. Simplify any expression inside the brackets first, if possible.
2. Look at the sign immediately before the bracket and apply the rule above.
3. Remove the brackets and combine like terms.

Quick reminder: always work from the innermost brackets outward.

WORKED_EXAMPLE

Example 1

(a) $a + (b + c)$

Since the sign before the bracket is $+$, the signs inside stay the same:
\[a + (b + c) = a + b + c\]

(b) $a + (b - c)$

Again the sign before the bracket is $+$, so:
\[a + (b - c) = a + b - c\]

(c) $2a - (3b - c)$

The sign before the bracket is $-$, so we change the signs inside:
\[2a - (3b - c) = 2a - 3b + c\]

Example 2

(a) $a + (a - b)$

\[a + (a - b) = a + a - b = 2a - b\]

(b) $6x + (3 - x)$

\[6x + (3 - x) = 6x + 3 - x = 5x + 3\]

(c) $(3a + 2) + (5a + a)$

\[(3a + 2) + (5a + a) = 3a + 2 + 5a + a = 9a + 2\]

Example 3

(a) $5x - (3y + z)$

The leading sign is $-$, so change the signs inside:
\[5x - (3y + z) = 5x - 3y - z\]

(b) $10d - (9c - 4d)$

\[10d - (9c - 4d) = 10d - 9c + 4d = 14d - 9c\]

(c) $(6x - y) - (4x - 5y)$

First bracket stays as is, second bracket is preceded by $-$:
\[(6x - y) - (4x - 5y) = 6x - y - 4x + 5y = 2x + 4y\]

(You could also factor the final result: $2(x + 2y)$.)

1. (a) Simplify \((12a - 7) + (4 - 5a)\)
answer: 7a - 3

2. (b) Simplify \((8x - 2y) - (9x - 4y)\)
answer: -x + 2y

3. Simplify \(12 - (7 - 2f) - (3 - f)\)
answer: 2 + 3f

4. Simplify \((8a - 5) + (14 - 3a)\)
answer: 5a + 9

Objective 3

Solving Word Problems with Algebraic Expressions

EXPLANATION
When a word problem gives the cost of one item and tells you how another item's cost relates to it, you can translate the words into an algebraic expression. Use a letter to stand for the unknown cost, then write the second item's cost in terms of that letter. Finally, combine the expressions to answer the question.

WORKED_EXAMPLE
Example
A pen costs P naira and a pencil costs 15 naira less than the pen.

(a) Cost of the pencil = $P-15$.

(b) Cost of a pen and a pencil = $P + (P-15) = 2P-15$.

The same steps work for any similar problem: identify the unknown, express the related cost, then add or subtract as required.

1. 1. A pen costs h naira and a crayon costs ₦20 less than a pen. (a) Find the cost of a crayon. (b) Find the total cost of a crayon and a pen.
answer: (a) ₦(h-20) (b) ₦(2h-20)

2. 2. Envelopes cost d naira per pack. A notepad costs ₦20 more than a pack of envelopes. Find the total cost of one pack of envelopes and one notepad.
answer: ₦(2d+20)

3. A pen and a pencil together cost x naira. If a pen costs y naira, find the cost of three pencils.
answer: ₦(3x-3y)

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	3	Scope	**grounded**
Explanation the principle, rule or relationship to review	3	Enacted content	**grounded**
Worked example a problem with its solution worked through	4	Enacted content	**grounded**

Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	14	Set work	**grounded**
Answer what the learner checks against	14	Enacted content	**grounded**